A negative answer to a question of Kierstead, Kostochka and Xiang on strongly equitable list coloring of forests

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Abstract

A forest T on n vertices is equitably k-colorable ($k \geq 3$) if and only if every vertex lies in an independent set of size at least $\lfloor n/k \rfloor$ (Chen–Lih; Miyata–Tokunaga–Kaneko). Kierstead, Kostochka and Xiang asked whether the same condition characterizes strongly equitable k-choosability of forests. We show that it does not: for every $k \geq 2$ there is a forest, and even a tree, on $2k^3 + k$ vertices that satisfies the independence condition (indeed is equitably k-colorable) but is not equitably k-choosable even in the weaker sense of Kostochka, Pelsmajer and West. The construction sharpens an example of Kaul, Mudrock and Wagstrom, and it generalizes to a family of necessary conditions for strongly equitable choosability of forests.

1 Introduction

All graphs are finite and simple. For a graph G and a vertex v, let $\alpha_v(G)$ denote the maximum size of an independent set of G containing v, and let $N(v)$, $N[v] = N(v) \cup \{v\}$ and $d(v) = |N(v)|$ be the neighborhood, closed neighborhood and degree of v. We write $[k] = \{1, \ldots, k\}$.

A proper vertex coloring is *equitable* if any two color classes differ in size by at most one. In an equitable k-coloring of an n-vertex graph every class has at least $\lfloor n/k \rfloor$ vertices, so the class of v witnesses $\alpha_v(G) \geq \lfloor n/k \rfloor$. For forests this obvious necessary condition is also sufficient.

Theorem 1 (Chen–Lih [2]; Miyata–Tokunaga–Kaneko [8]; see Chang [1])**.** *Let $k \geq 3$. A forest T on n vertices is equitably k-colorable if and only if $\alpha_v(T) \geq \lfloor n/k \rfloor$ for every vertex v.*

We call the condition in Theorem 1 the *independence condition*.

List versions

A *k-list assignment* for G is a map L from $V(G)$ to sets of colors with $|L(v)| = k$ for all v, and an *L-coloring* is a proper coloring f with $f(v) \in L(v)$ for all v. Kostochka, Pelsmajer and West [7] call an L-coloring *equitable* if every color class has at most $\lceil |G|/k \rceil$ vertices, and G *equitably k-choosable* if it has an equitable L-coloring for every k-list assignment L. Note that an equitable L-coloring may use more than k colors and may have very small classes; only an upper bound on class sizes is imposed.

Kierstead, Kostochka and Xiang [4, 5] introduced a stronger notion. Write $|G| = kq + r$ with $q \in \mathbb{N}$ and $1 \leq r \leq k$, and set $|G| \operatorname{mod}^* k := r$ (so $|G| \operatorname{mod}^* k = k$ when k divides $|G|$). A color class is *full* if it has exactly $\lceil |G|/k \rceil$ vertices. An L-coloring is *strongly equitable* (SE) if no class has more than $\lceil |G|/k \rceil$ vertices and at most $|G| \bmod^* k$ classes are full; G is SE *k-choosable* if it has an SE L-coloring for every k-list assignment L. Two features of this definition matter here.

- Every SE L-coloring is an equitable L-coloring in the sense of [7]; hence SE k-choosable graphs are equitably k-choosable.
- If all lists equal $[k]$, then the SE L-colorings are exactly the equitable k-colorings: with $r = |G| \bmod^* k$ there are r classes of size $q+1$ and $k-r$ of size q. Hence SE k-choosable graphs are equitably k-colorable, and in particular SE k-choosable forests satisfy the independence condition.

In their survey [6, Question 13], Kierstead, Kostochka and Xiang asked whether Theorem 1 extends to this setting.

Question 2 ([6])**.** Is it true that for each $k \geq 3$, an n-vertex forest T is SE k-choosable if and only if $\alpha_v(T) \geq \lfloor n/k \rfloor$ for every vertex v?

The question is natural: by the second feature above, the "only if" direction holds, and for ordinary equitable coloring the independence condition is the only obstruction. We show that in the list setting the "if" direction fails for every k, even for trees, and even for the weaker notion of [7]. The failure is by the smallest possible margin (Remark 4), which suggests that a modified characterization may still exist; see Section 4.

Theorem 3. *Let $k \geq 2$, $q = 2k^2 + 1$ and $n = kq = 2k^3 + k$. There is a forest F_k and a tree T_k, each on n vertices, such that*

(i) *F_k and T_k have equitable k-colorings; in particular $\alpha_v \geq q = \lfloor n/k \rfloor$ for every vertex v;*

(ii) *F_k and T_k are not equitably k-choosable in the sense of [7], and hence not* SE *k-choosable.*

Consequently the answer to Question 2 is negative for every $k \geq 3$.

The forest F_k is a disjoint union of two stars. Kaul, Mudrock and Wagstrom [3, Proposition 22] showed that $K_{1,(k-1)(k^3-k+2)} + K_{1,k^3}$ is not equitably k-choosable; their example was constructed to show that a sufficient condition in [3] cannot be relaxed, and Question 2 was posed later. Read in the present context, that forest also satisfies the independence condition and thus already gives a negative answer. Our F_k is the smallest member of the same family of examples (Remark 4). In Section 3 the argument is generalized to a family of necessary conditions for SE k-choosability of forests, and Section 4 collects open questions.

2 The construction

Throughout this section $k \geq 2$, $q = 2k^2 + 1$ and $n = kq$. Since k divides n, we have $\lfloor n/k \rfloor = \lceil n/k \rceil = q$, and an L-coloring is equitable in the sense of [7] exactly when every color is used at most q times.

The forest

Let F_k be the disjoint union of two stars: one with center u_0 and leaf set A, where $|A| = (k-1)q - 1$, and one with center w_0 and leaf set B, where $|B| = 2k^2$. Then

$$|F_k| = 2 + \big((k-1)q - 1\big) + 2k^2 = (k-1)q + q = kq = n.$$

Index the leaves of w_0 by triples, $B = \{b_{c,d,t} : c \in [k],\ d \in \{k+1, \dots, 2k\},\ t \in \{1, 2\}\}$.

Equitable colorability

The set $\{u_0\} \cup B$ is independent and has $1 + 2k^2 = q$ vertices; the set $\{w_0\} \cup A$ is independent and has $(k-1)q$ vertices. Using $\{u_0\} \cup B$ as one class and splitting $\{w_0\} \cup A$ into $k-1$ classes of size q gives an equitable k-coloring of F_k. This proves (i) of Theorem 3 for F_k.

The list assignment

Define a k-list assignment L by

$$L(u_0) = L(a) = [k] \quad (a \in A), \qquad L(w_0) = \{k+1, \dots, 2k\}, \qquad L(b_{c,d,t}) = \big([k] \setminus \{c\}\big) \cup \{d\}.$$

No equitable L-coloring

Suppose f is an L-coloring of F_k in which every color is used at most q times. Let $c = f(u_0) \in [k]$ and $d = f(w_0) \in \{k+1, \dots, 2k\}$. Consider the set

$$S = A \cup \{b_{c,d,1},\, b_{c,d,2}\}, \qquad |S| = (k-1)q + 1.$$

Every $a \in A$ is adjacent to u_0 and has list $[k]$, so $f(a) \in [k] \setminus \{c\}$. Each of $b_{c,d,1}, b_{c,d,2}$ is adjacent to w_0, so it does not receive d, and its remaining colors are $[k] \setminus \{c\}$. Thus f maps the $(k-1)q+1$ vertices of S into the $k-1$ colors of $[k] \setminus \{c\}$, and by the pigeonhole principle some color is used more than q times, a contradiction. Hence F_k is not equitably k-choosable, and, since every SE L-coloring is an equitable L-coloring, not SE k-choosable. This proves (ii) for F_k.

The tree

Fix $a_0 \in A$ and $b_0 \in B$ and let $T_k = F_k + a_0 b_0$. Adding an edge cannot create an L-coloring, so T_k is not equitably k-choosable. For (i), $\{u_0\} \cup B$ and $\{w_0\} \cup A$ are still independent in T_k (neither contains both a_0 and b_0), and the equitable k-coloring above places a_0 and b_0 in different classes, so it is a proper equitable k-coloring of T_k. This completes the proof of Theorem 3. $\square$

Remark 4. In the construction, the leaves of u_0 fill the $k-1$ colors $[k] \setminus \{c\}$ up to a *slack* of exactly one vertex, and for each of the k^2 possible pairs $(f(u_0), f(w_0))$ two leaves of w_0 compete for that single slot. The example of [3, Proposition 22] is the same mechanism with slack $k-1$ and k competing leaves per pair, which costs $n = k(k^3 - k + 3)$ vertices instead of $2k^3 + k$ (81 instead of 57 for $k = 3$). Slack zero is impossible: if $|A| = (k-1)q$ then $|B| = q - 2$ and $\alpha_{u_0} = q - 1$, which violates the independence condition. This is precisely the obstruction that makes Theorem 1 true; the list version escapes it by a single vertex, so the independence condition fails to be sufficient by the smallest possible margin.

3 A general obstruction

The proof of Theorem 3 only used a count of vertices forced into $k-1$ colors. We record the general form of this count.

Let T be a forest on n vertices and write $n = kq + r'$ with $0 \le r' \le k-1$, so that $q = \lfloor n/k \rfloor$. In an SE L-coloring, any $k-1$ colors together carry at most $n - q$ vertices: if $r' = 0$ every class has at most q vertices, and if $r' \ge 1$ every class has at most $q+1$ vertices with at most r' classes of size $q+1$, so the bound is $(k-1)q + r' = n - q$ in both cases. In an equitable L-coloring in the sense of [7] the corresponding bound is $(k-1)\lceil n/k \rceil$.

Proposition 5. *Let T be a forest on n vertices, $q = \lfloor n/k \rfloor$, let $u \in V(T)$, let $W \subseteq V(T) \setminus \{u\}$, and let $Y_w \subseteq N(w) \setminus N[u]$ $(w \in W)$ be pairwise disjoint. If*

$$d(u) - |W \cap N(u)| + \sum_{w \in W} \left\lfloor \frac{|Y_w|}{k^2} \right\rfloor \; > \; n - q, \tag{1}$$

then T is not SE *k-choosable. If (1) holds with $n - q$ replaced by $(k-1)\lceil n/k \rceil$, then T is not equitably k-choosable in the sense of [7].*

Proof. Let $D = \{k+1, \ldots, 2k\}$. Define $L(u) = [k]$, $L(x) = [k]$ for $x \in N(u) \setminus W$, and $L(w) = D$ for $w \in W$. For each $w \in W$ split Y_w into k^2 parts $Y_{w,c,d}$ $(c \in [k],\, d \in D)$, each of size at least $\lfloor |Y_w|/k^2 \rfloor$, and give every $y \in Y_{w,c,d}$ the list $([k] \setminus \{c\}) \cup \{d\}$. All other lists are arbitrary k-sets. Let f be an SE L-coloring and put $c = f(u)$. Every $x \in N(u) \setminus W$ is colored from $[k] \setminus \{c\}$. For $w \in W$ let $d = f(w) \in D$; every $y \in Y_{w,c,d}$ is adjacent to w, cannot receive d, and is therefore colored from $[k] \setminus \{c\}$. The number of these vertices is at least the left-hand side of (1), which exceeds the capacity $n - q$ of $k - 1$ colors. The second statement is identical with the other capacity. $\square$

Theorem 3 is the case $W = \{w_0\}$, $Y_{w_0} = B$, $|B| = 2k^2$, $d(u_0) = (k-1)q - 1$, where the left-hand side equals $(k-1)q + 1 = n - q + 1$. With $W = \emptyset$, condition (1) reads $d(u) > n - q$; since $\alpha_u \leq n - d(u)$, this is already excluded by the independence condition. Thus Proposition 5 adds new information only when some vertex w has at least k^2 neighbors outside $N[u]$.

4 Concluding remarks and open questions

Verification

The forest F_k, the list assignment L, the independence condition and the non-existence of an SE L-coloring have been formalized for all $k \geq 2$ and machine-checked in Lean 4 with Mathlib (file `Q13lean/Counterexample.lean` in the accompanying repository). Independently, an exact flow-based program confirms Theorem 3 for $k = 3, 4$ and the example of [3] for $k = 3, 4$.

Small forests

An exhaustive computer search found no forest satisfying the independence condition and admitting a defeating k-list assignment in the following ranges, where the palette is the set of colors from which all lists are drawn.

k	number of vertices	palette size
3	≤ 12	4
3	≤ 10	5
4	≤ 11	5
5	≤ 10	6

Because a defeating assignment may need a larger palette (L above uses $2k$ colors), these searches do not yield a lower bound on the size of a smallest counterexample.

Question 6. For $k = 3$, what is the smallest number of vertices of a forest that satisfies the independence condition but is not SE 3-choosable? It is at most 57.

A corrected characterization?

The independence condition together with the family of conditions "(1) fails for all choices of u, W, (Y_w)" is necessary for SE k-choosability of a forest. We do not know whether it is sufficient. A first test case is the class of disjoint unions of two stars, where even equitable k-choosability in the sense of [7] has not been characterized [3, Question 8].

Question 7. Is a forest T on n vertices SE k-choosable whenever $\alpha_v(T) \geq \lfloor n/k \rfloor$ for all v and (1) fails for all u, W and (Y_w)? In particular, is T SE k-choosable whenever the independence condition holds and $\Delta(T) < k^2$?

For comparison, Kostochka, Pelsmajer and West [7] proved that every forest is equitably k-choosable (in the sense of [7]) when $k \geq 1 + \Delta/2$, and that this bound is tight (as quoted in [3]). No such degree bound can hold for SE k-choosability without the independence condition, since SE k-choosable graphs are equitably k-colorable; whether the independence condition suffices for SE k-choosability of forests with $\Delta \leq 2k - 2$ is a natural special case of the question above. By Proposition 5, the new list obstructions appear only at degree k^2 and beyond.

References

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